

University Preparatory Program (Mathematics) Improper Integrals



Engineering Mathematics Team

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2. Improper Integrals

Learning Objectives

Upon completion of this lesson, you should be able to:

1. Calculate limit of a function using algebraic/trigonometric simplification
2. Calculate limit of a function using the Squeeze Theorem or the l'Hôpital rule
3. Calculate improper integrals of the 1st and 2nd kinds

2.1 Limits of a Function

In the study of the limit of a function $y = f(x)$, we would like to observe the behavior of the output value y when the input value x approaches a particular number, say c . In mathematical notation, we would write: $y \rightarrow ?$ as $x \rightarrow c$. Consider the function

$$f(x) = \frac{x^2 - 4}{x - 2}.$$

Obviously, $f(2)$ does not exist; notice the 'hole' on its graph in Figure 2.1.1. Simplifying algebraically, we write

$$f(x) = x + 2 \text{ for } x \neq 2.$$

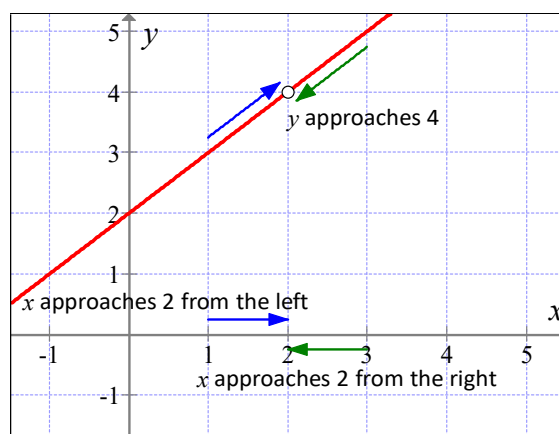


Figure 2.1.1. Limit of $y = \frac{x^2 - 4}{x - 2}$ as $x \rightarrow 2$

From the graph in Figure 1, we observe that as x approaches 2 from either direction (left or right), the output y approaches 4. In mathematical language, we say that as x tends to 2, the **limit** of f is 4, and we use the notation: $y \rightarrow 4$ as $x \rightarrow 2$, or more formally,

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = 4.$$

Notice the two directions x approaches 2: from the left and from the right. The mathematical notations for these are as follows.

x approaches c from the left: $x \rightarrow c^-$

x approaches c from the right: $x \rightarrow c^+$

Each of this direction defines a **one-sided limit**: left limit or right limit.

In Figure 1, we observe that “ y approaches 4 as x approaches 2 from the left”. In mathematical language, we write: $y \rightarrow 4$ as $x \rightarrow 2^-$, or more formally

$$\lim_{x \rightarrow 2^-} \frac{x^2 - 4}{x - 2} = 4.$$

Similarly, from the graph, we also observe that

$$\lim_{x \rightarrow 2^+} \frac{x^2 - 4}{x - 2} = 4.$$

Notice that in this example, as x tends to 2, the left and right limits of f are equal. We now present an informal definition of limit.

Definition 2.1.1

As x tends to c , the **limit** of $f(x)$ is L if both its left and right limits equal L . In other words,

$$\begin{aligned} \lim_{x \rightarrow c} f(x) &= L \\ \text{if } \lim_{x \rightarrow c^-} f(x) &= \lim_{x \rightarrow c^+} f(x) = L. \end{aligned}$$

For many elementary and trigonometric functions, we find the limit by evaluating the function at that particular input value. This is summarized as follows.

Limit Rule #1

Let $f(x)$ be any of the following functions: constant, polynomial, exponential, logarithmic, rational, root, trigonometric, and inverse trigonometric. If c is in the domain of f , then

$$\lim_{x \rightarrow c} f(x) = f(c).$$

Example 2.1.2

- (a) $\lim_{x \rightarrow 125} \sqrt{x} = \sqrt{125}$
- (b) $\lim_{x \rightarrow -1} (3x^3 - 2x + 5) = 3(-1)^3 - 2(-1) + 5 = 4$
- (c) $\lim_{x \rightarrow 2} e^{-x} = e^{-2} \approx 0.1353$
- (d) $\lim_{x \rightarrow 0} \frac{1}{x+1} = \frac{1}{0+1} = 1$
- (e) $\lim_{x \rightarrow \pi} \cos x = \cos \pi = -1$
- (f) $\lim_{x \rightarrow 1} \tan^{-1} x = \tan^{-1}(1) = \frac{\pi}{4}$

Limit Rule #2

Suppose both limits $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ are finite, say

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M.$$

Then basic algebraic operations (add, subtract, product, quotient, power) can be applied to these limits, that is,

$$\lim_{x \rightarrow c} [f(x) \pm g(x)] = L \pm M$$

$$\lim_{x \rightarrow c} [f(x)g(x)] = LM$$

$$\lim_{x \rightarrow c} \left[\frac{f(x)}{g(x)} \right] = \frac{L}{M} \quad \text{provided } M \neq 0$$

$$\lim_{x \rightarrow c} [f(x)]^n = L^n \quad \text{provided } L^n \text{ exists}$$

Example 2.1.3

$$\lim_{x \rightarrow 1} \frac{x^3 - 4x}{x - 2} = \frac{1 - 4}{1 - 2} = 3$$

We are particularly interested in evaluating two kinds of limits:

1. Limit of a function that involves a division with denominator approaches 0
2. Limit of a function as x increasing without bound, i.e. $x \rightarrow \infty$

The following examples show the case where the limit involves a division with the denominator approaches 0. In such cases, algebraic or trigonometric simplification is needed before applying Limit Rules #1 or #2.

Example 2.1.4

$$\lim_{x \rightarrow 2} \frac{x^3 - 4x}{x - 2} = \lim_{x \rightarrow 2} \frac{x(x^2 - 4)}{x - 2} = \lim_{x \rightarrow 2} \frac{x(x + 2)(x - 2)}{x - 2} = \lim_{x \rightarrow 2} x(x + 2) = 2(2 + 2) = 8$$

Example 2.1.5

$$\lim_{x \rightarrow 1} \frac{x - 1}{1 - \sqrt{x}} = \lim_{x \rightarrow 1} \frac{(x - 1)(1 + \sqrt{x})}{(1 - \sqrt{x})(1 + \sqrt{x})} = \lim_{x \rightarrow 1} \frac{(x - 1)(1 + \sqrt{x})}{1 - x} = \lim_{x \rightarrow 1} -(1 + \sqrt{x}) = -2$$

Example 2.1.6

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{\sin x} = \lim_{x \rightarrow 0} \frac{2 \sin x \cos x}{\sin x} = \lim_{x \rightarrow 0} 2 \cos x = 2 \cos 0 = 2$$

The following Limit Rule #3 involves the **infinity** symbols, ∞ , to denote increasing values without bound.

Limit Rule #3

To calculate the limit of a rational function as x tends to infinity, i.e. $x \rightarrow \infty$, we use the following principle

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

In other words, the limit of a quotient is zero when the numerator is constant and the denominator is increasing without bound.

Example 2.1.7

$$\lim_{x \rightarrow \infty} \frac{x^2 + 1}{1 - 2x^2} = \lim_{x \rightarrow \infty} \frac{(x^2 + 1) \div x^2}{(1 - 2x^2) \div x^2} = \lim_{x \rightarrow \infty} \frac{1 + \frac{1}{x^2}}{\frac{1}{x^2} - 2} = \frac{1 + 0}{0 - 2} = -0.5.$$

Limit Rule #4

As x tends to infinity, the following are the limits of some functions.

$$\lim_{x \rightarrow \infty} e^x = \infty$$

$$\lim_{x \rightarrow \infty} \ln x = \infty$$

$$\lim_{x \rightarrow \infty} \tan^{-1} x = \frac{\pi}{2}$$

$$\lim_{x \rightarrow -\infty} \tan^{-1} x = -\frac{\pi}{2}$$

Example 2.1.8

$$(a) \quad \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$

$$(b) \quad \lim_{x \rightarrow \infty} \frac{x^2 + 1}{x - 1} = \lim_{x \rightarrow \infty} \left(x + 1 + \frac{2}{x - 1} \right) = \infty$$

$$(c) \quad \lim_{x \rightarrow \infty} \tan^{-1}(x^2 + 1) = \frac{\pi}{2}$$

WARNING!

Be careful in dealing with infinity. Since ∞ is not a number, expressions such as $\infty - \infty$, $\frac{\infty}{\infty}$, or 0^∞ are called **indeterminate forms**. These forms are to be treated differently because the usual algebraic operations are NOT applicable for ∞ .

The Limit Rule #5 involves unbounded or infinite limits. The symbol $-\infty$ denotes decreasing values without bound.

Limit Rule #5

As $x \rightarrow c$, the limit of f is unbounded if either $y \rightarrow \infty$ or $y \rightarrow -\infty$, i.e.

$$\lim_{x \rightarrow c} f(x) = \infty \quad \text{or} \quad \lim_{x \rightarrow c} f(x) = -\infty$$

The following are common unbounded limits.

$$\lim_{x \rightarrow 0^+} \ln x = -\infty$$

$$\lim_{x \rightarrow a} \frac{1}{(x-a)^2} = \infty$$

$$\lim_{x \rightarrow a^-} \frac{1}{x-a} = -\infty$$

$$\lim_{x \rightarrow \frac{\pi}{2}^-} \tan x = \infty$$

$$\lim_{x \rightarrow a^+} \frac{1}{x-a} = \infty$$

$$\lim_{x \rightarrow \frac{\pi}{2}^+} \tan x = -\infty$$

Example 2.1.9

(a) $\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$ (note: the limit $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist; it is NOT ∞)

(b) $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$

Sometimes, we require more than simple algebraic or trigonometric simplification to calculate a limit. We shall learn two techniques in calculating limits: the Squeeze Theorem and the l'Hôpital Rule.

Theorem 2.1.10 (The Squeeze Theorem)

Suppose $P(x) \leq f(x) \leq Q(x)$ for all x in an interval I . If c is in the interval I , and

$$\lim_{x \rightarrow c} P(x) = \lim_{x \rightarrow c} Q(x) = L, \text{ then } \lim_{x \rightarrow c} f(x) = L.$$

The Squeeze theorem is valid for $x \rightarrow \infty$ if the inequality $P(x) \leq f(x) \leq Q(x)$ holds for all $x > K$ for some positive number K .

Example 2.1.11

Calculate $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$.

Since $-1 \leq \sin x \leq 1$ for all $x > 0$, we have $-\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$ for all $x > 0$.

Now, since $\lim_{x \rightarrow \infty} \left(-\frac{1}{x}\right) = \lim_{x \rightarrow \infty} \left(\frac{1}{x}\right) = 0$, by the squeeze theorem, we conclude that $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$.

Theorem 2.1.12 (The l'Hôpital Rule)

Suppose f and g are differentiable functions of x . If, as x tends to c , the limits of f and g are both zero or both unbounded, i.e. $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ is of *indeterminate form* $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$, then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

Example 2.1.13

Calculate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

Since $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{0}{0}$ (indeterminate), we apply l'Hôpital rule and obtain

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(\sin x)}{\frac{d}{dx}(x)} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = \cos 0 = 1.$$

The following examples show the use of the l'Hôpital rule for $x \rightarrow \infty$.

Example 2.1.14

Calculate $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$.

Since $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \frac{\infty}{\infty}$ (indeterminate), we apply l'Hôpital rule and obtain

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x)} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0.$$

2.2 Improper Integrals

Improper integrals of the 1st kind are integrals of the form

$$\int_a^\infty f(x) dx \text{ or } \int_{-\infty}^b f(x) dx \text{ or } \int_{-\infty}^\infty f(x) dx$$

A limit process will be applied to calculate improper integrals.

Definition 2.2.1 (Improper Integrals of the 1st Kind)

$$\int_a^\infty f(x) dx = \lim_{N \rightarrow \infty} \int_a^N f(x) dx \quad \text{if } f \text{ is integrable on } [a, N] \text{ for all } N > a$$

$$\int_{-\infty}^b f(x) dx = \lim_{M \rightarrow -\infty} \int_M^b f(x) dx \quad \text{if } f \text{ is integrable on } [M, b] \text{ for all } M < b$$

$$\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^\infty f(x) dx \quad \text{if both limits on the RHS exist}$$

An improper integral is said to be **convergent** if the limit exists; otherwise it is **divergent**.

Example 2.2.2

$$\int_1^\infty \frac{1}{x^2} dx = \lim_{N \rightarrow \infty} \int_1^N \frac{1}{x^2} dx = \lim_{N \rightarrow \infty} \left[-\frac{1}{x} \right]_1^N = \lim_{N \rightarrow \infty} \left[1 - \frac{1}{N} \right] = 1.$$

Example 2.2.3

$$\begin{aligned} \int_{-\infty}^\infty \frac{1}{1+x^2} dx &= \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^\infty \frac{1}{1+x^2} dx \\ &= \lim_{M \rightarrow -\infty} \int_M^0 \frac{1}{1+x^2} dx + \lim_{N \rightarrow \infty} \int_0^N \frac{1}{1+x^2} dx \\ &= \lim_{M \rightarrow -\infty} [\tan^{-1} 0 - \tan^{-1} M] + \lim_{N \rightarrow \infty} [\tan^{-1} N - \tan^{-1} 0] \\ &= \left(0 - \left(-\frac{\pi}{2} \right) \right) + \left(\frac{\pi}{2} - 0 \right) = \pi. \end{aligned}$$

Example 2.2.4

$$\begin{aligned} \int_0^\infty \frac{x}{e^x} dx &= \lim_{N \rightarrow \infty} \int_0^N x e^{-x} dx = \lim_{N \rightarrow \infty} \left(\left[-\frac{x}{e^x} \right]_0^N + \int_0^N e^{-x} dx \right) \\ &= \lim_{N \rightarrow \infty} \left[-\frac{x}{e^x} - \frac{1}{e^x} \right]_0^N = \lim_{N \rightarrow \infty} \left[-\frac{N+1}{e^N} + \frac{0+1}{e^0} \right] \\ &= 1 - \lim_{N \rightarrow \infty} \left[\frac{N+1}{e^N} \right] \\ &= 1 - \lim_{N \rightarrow \infty} \left[\frac{1}{e^N} \right] \quad \text{by l'Hôpital rule} \\ &= 1. \end{aligned}$$

Improper integrals of the 2nd kind are integrals of the form

$$\int_a^b f(x) dx, \text{ where } f \text{ is unbounded at some point } c \text{ in the interval } [a, b]$$

Recall that a function f is unbounded at c if $\lim_{x \rightarrow c} f(x) = \infty$ or $-\infty$. We calculate improper integrals of the 2nd kind as follows.

Definition 2.2.5 (Improper Integrals of the 2nd Kind)

The value (if exists) of an improper integral $\int_a^b f(x) dx$ equals to

$$\lim_{t \rightarrow a^+} \int_t^b f(x) dx \quad \text{if } f \text{ is unbounded at } a, \text{ but integrable on } [t, b] \text{ for all } a < t \leq b$$

$$\lim_{t \rightarrow b^-} \int_a^t f(x) dx \quad \text{if } f \text{ is unbounded at } b, \text{ but integrable on } [a, t] \text{ for all } a \leq t < b$$

$$\lim_{t \rightarrow c^-} \int_a^t f(x) dx + \lim_{u \rightarrow c^+} \int_u^b f(x) dx \quad \text{if } f \text{ is unbounded at some point } c \text{ in the interval } (a, b), \text{ but integrable on } [a, t] \text{ and } [u, b] \text{ for all } a \leq t < c \text{ and } c < u \leq b.$$

Example 2.2.6

$$\begin{aligned} \int_0^1 \frac{1}{\sqrt{1-x}} dx &= \lim_{t \rightarrow 1^-} \int_0^t \frac{1}{\sqrt{1-x}} dx \\ &= \lim_{t \rightarrow 1^-} [-2\sqrt{1-x}]_0^t = -2 \lim_{t \rightarrow 1^-} [\sqrt{1-t} - \sqrt{1}] \\ &= -2(0 - 1) = 2. \end{aligned}$$

Example 2.2.7

$$\begin{aligned} \int_0^1 \ln x dx &= \lim_{t \rightarrow 0^+} \int_t^1 \ln x dx = \lim_{t \rightarrow 0^+} [x \ln x - x]_t^1 \\ &= \lim_{t \rightarrow 0^+} [(0 - 1) - (t \ln t - t)] = \lim_{t \rightarrow 0^+} \left[-1 - \frac{\ln t}{\frac{1}{t}} + t \right] \\ &= -1 - \lim_{t \rightarrow 0^+} \left[\frac{\frac{1}{t}}{-\frac{1}{t^2}} \right] + 0 \quad \text{by l'Hôpital rule since } \lim_{t \rightarrow 0^+} \frac{\ln t}{\frac{1}{t}} = \frac{-\infty}{\infty} \\ &= -1 + \lim_{t \rightarrow 0^+} t = -1 + 0 = -1. \end{aligned}$$

2.3 Exercises

1. Find each of the following limits.

(a) $\lim_{t \rightarrow 1^-} \sqrt{t}$

(f) $\lim_{x \rightarrow \infty} \frac{x+2}{x+3}$

(b) $\lim_{t \rightarrow 1^-} \sin^{-1} t$

(g) $\lim_{x \rightarrow \infty} \ln \left| \frac{x+2}{x+3} \right|$

(c) $\lim_{t \rightarrow 1^+} \ln|1-t|$

(h) $\lim_{x \rightarrow \infty} e^{-x^2}$

(d) $\lim_{x \rightarrow \infty} (\tan^{-1} x)^2$

(i) $\lim_{x \rightarrow -\infty} x e^x$

(e) $\lim_{x \rightarrow \infty} \tan^{-1}(\sqrt{x^2-1})$

(j) $\lim_{x \rightarrow \infty} e^{-x} \sin x$

2. A function f is said to be **continuous** at $x = x_0$ if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

Sketch the graph of each of the following functions, and determine whether it is continuous at the specified point by calculating its left and right limits at that point.

(a) $f(x) = |x|$ at $x = 0$

(c) $f(x) = |\sin x|$ at $x = \pi$

(b) $f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$ at $x = 0$

(d) $f(x) = \begin{cases} x & \text{if } x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$ at $x = 1$

3. The **derivative** $f'(x)$ of a function f is defined as $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$.

Without applying l'Hôpital rule, use the definition above to find the derivative of each of the following functions.

(a) $f(x) = \sin x$

(c) $f(x) = |x|$ at $x = 0$

(b) $f(x) = \cos x$

(d) $f(x) = \begin{cases} x & \text{if } x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$ at $x = 1$

Note: for (a) and (b), use the fact that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ and $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$

4. Calculate each of the following improper integrals of the 1st kind.

(a) $\int_0^{\infty} \frac{1}{x^2 + 1} dx$

(f) $\int_{-1}^{\infty} \frac{1}{x^2 + 5x + 6} dx$

(b) $\int_1^{\infty} \frac{1}{x^{1.001}} dx$

(g) $\int_{-\infty}^0 xe^x dx$

(c) $\int_2^{\infty} \frac{1}{(x-1)^2} dx$

(h) $\int_0^{\infty} 2e^{-x} \sin x dx$

(d) $\int_0^{\infty} \frac{16 \tan^{-1} x}{x^2 + 1} dx$

(i) $\int_{-\infty}^{\infty} 2xe^{-x^2} dx$

(e) $\int_2^{\infty} \frac{2}{t^2 - 1} dt$

(j) $\int_{-\infty}^{\infty} \frac{x}{(x^2 + 4)^{3/2}} dx$

5. Calculate each of the following improper integrals of the 2nd kind.

(a) $\int_0^1 \frac{1}{\sqrt{x}} dx$

(f) $\int_0^{\pi/2} (\sec x - \tan x) dx$

(b) $\int_0^1 \frac{1}{\sqrt{1-x^2}} dx$

(g) $\int_{-1}^1 \frac{1}{x^{2/3}} dx$

(c) $\int_0^1 \frac{x+1}{\sqrt{x^2+2x}} dx$

(h) $\int_1^2 \frac{1}{1-x} dx$

(d) $\int_0^1 x \ln x dx$

(i) $\int_0^2 \frac{1}{\sqrt{|x-1|}} dx$

(e) $\int_0^1 \frac{4x}{\sqrt{1-x^4}} dx$

(j) $\int_1^{\infty} \frac{1}{x\sqrt{x^2-1}} dx$

6. The **Laplace transform** of a function f , denoted by $\mathcal{L}(f)$, is a function of s defined as

$$\mathcal{L}(f) = \int_0^{\infty} e^{-st} f(t) dt$$

Find the Laplace transform of each of the following functions.

(a) $f(t) = 1$

(c) $f(t) = t$

(b) $f(t) = e^{at}$

(d) $f(t) = \sin t$

